

10 Acoustic Characteristics of Speech Sounds

This chapter discusses the primary acoustic characteristics of a variety of speech sounds and illustrates them by means of spectrograms and spectra. We start with vowels and subsequently discuss consonants.

10.1 Vowels

Vowels are produced with a relatively open vocal tract and the airstream is not severely impeded. The resulting acoustic signal is therefore relatively loud. In addition, vowels are usually produced with vocal fold vibration. The primary acoustic characteristic of vowels is the location of the formant frequencies, specifically, the first three formants (F1–F3). As discussed in Section 9.5.1, the shape of the vocal tract determines the location of the formant frequencies. Changes in the position of the articulators will modify the shape of the vocal tract and, as a result, the location of the formant frequencies. Since the same formant frequencies can be generated with a variety of articulatory positions, formant frequency location is a critical determinant of vowel quality rather than the positions of the articulators. For a given speaker or for a group of speakers with the same vocal tract length, each vowel is associated with a distinct acoustic formant frequency pattern. As an example, Figure 10.1 shows spectrograms and linear predictive coding (LPC) spectra of the vowels [i] and [a] spoken by a male English speaker.

As on any spectrogram, frequency is plotted along the vertical axis, and time along the horizontal axis. Intensity is represented by the darkness of the display. The darker a particular area, the greater its intensity. The spectrograms of the two vowels have several things in common. They are both characterized by the presence of a number of dark bands along the frequency scale. These dark bands correspond to the formant frequencies that reflect the resonances of the vocal tract and appear as “peaks” in the LPC spectrum.

The crucial difference between the two vowels is of course the exact location of the formant frequencies. As indicated in Figure 10.1, the vowel [i] has a first formant frequency around 240 Hz, a second formant frequency around 2,450 Hz,

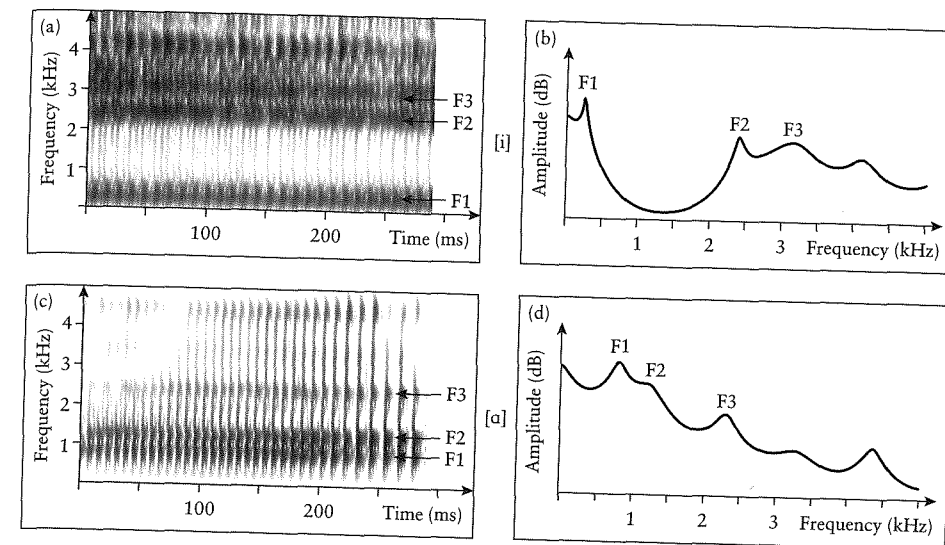


Figure 10.1 (a, c) Spectrograms and (b, d) LPC spectra of the English vowels [i] and [a] produced by a male speaker.

and a third formant frequency around 3,200 Hz. In contrast, the vowel [a] has a first formant frequency around 810 Hz, a second formant frequency around 1,250 Hz, and a third formant frequency around 2,400 Hz. This difference reflects the different vocal tract shapes involved in the production of the two vowels. Figure 10.2 shows the stylized frequencies of the first three formants for all monophthongal vowels of American English, averaged over a group of 50 male speakers (Hillenbrand et al. 1995).

The vowels along the horizontal axis are organized from front to back. In addition, the front vowels are ranked in descending order of height, from high [i] to low [æ], and the back vowels in ascending order of height, from low [a] to high [u]. When the vowels are arranged this way, it is apparent that the two

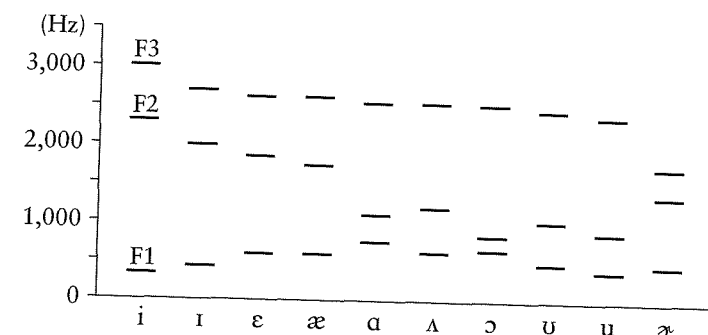


Figure 10.2 Average and stylized formant frequencies (F1–F3) of the monophthongal vowels of American English produced by 50 male speakers.

major dimensions for describing vowels, height and backness, have clear acoustic correlates. Vowel height is inversely correlated with the frequency of the first formant: the higher the vowel (and the higher the tongue position), the lower the F1. This is true for both the front and back vowels. Moreover, vowels of comparable height, such as [i] and [u], have comparable first formant frequencies. Vowel backness is reflected in the frequency of the second formant frequency, or more precisely, in the distance between the first and second formant frequencies. It is clear from Figure 10.2 that the difference $F2-F1$ is relatively large for the front vowels and relatively small for the back vowels.

As vowel quality changes, the frequency of the third formant does not change nearly as much as that of F1 and F2, with the possible exception of the vowel [i], for which F3 is quite high. Overall, then, F3 does not provide much information about the quality of the English vowels, which is why many acoustic descriptions of these vowels consist only of the first two formant frequencies. However, there are languages in which F3 does provide an important cue to vowel quality. For example, languages such as Dutch, French, German, and Swedish have both front unrounded and front rounded vowels. The lengthening of the vocal tract due to lip rounding lowers all formants (see Section 9.5.1), and moves F3 close to F2, which distinguishes between front unrounded and front rounded vowels, as illustrated in Figure 10.3 for German.

Formant frequencies higher than F3 are not considered important cues to the identity of a vowel because they hardly vary as a function of vowel quality. Instead, for vowels, the frequencies of higher formants such as F4 and F5 seem more speaker-specific and may therefore provide information about the identity of the speaker rather than the vowel itself. Therefore, for vowels, F1-F3 provide acoustic cues

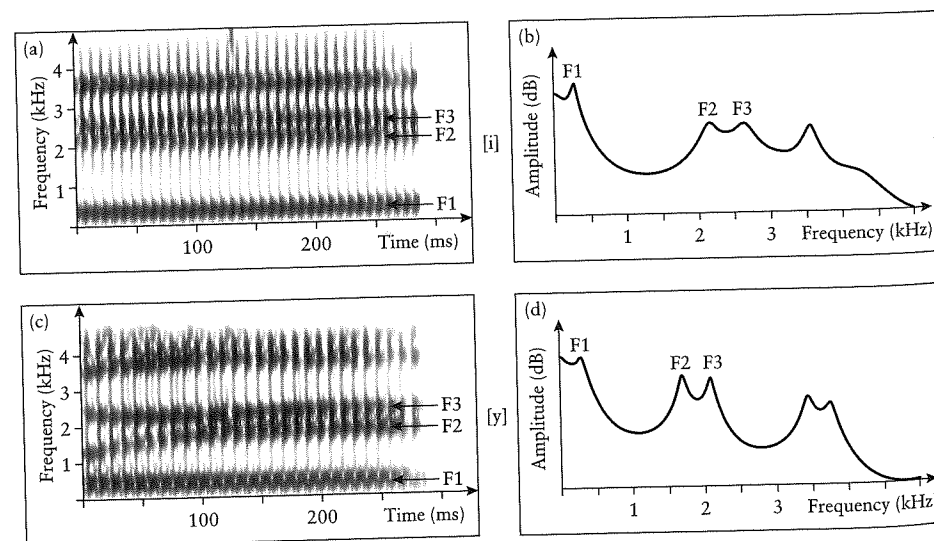


Figure 10.3 Spectrograms and LPC spectra of (a, b) the unrounded high front vowel [i] and (c, d) its rounded counterpart [y] spoken by a native male speaker of German.

to vowel quality. An acoustic cue consists of one or more acoustic properties that are considered to provide unique information about the identity of a particular segment such as the height, frontness, or rounding of a vowel.

The production of nasal and nasalized vowels requires two resonators, the oral and the nasal cavity. The complex interaction between these two resonators and the heavy damping of the nasal cavity result in several differences between oral and nasal(ized) vowels. Compared to an oral vowel, a nasal vowel typically shows greater formant bandwidths, lower overall amplitude, a low-frequency nasal formant, and one or more anti-formants (see Section 10.2.4 for a more detailed discussion of the acoustic correlates of nasalization). Figure 10.4 shows spectrograms of the oral vowel [a] and the nasal vowel [ã] spoken by the same speaker. Notice in Figure 10.4b the increased energy in the second harmonic around 200 Hz and the dip in the spectrum around 450 Hz.

Duration can also be a cue to a vowel's identity. Many languages have vowel distinctions that are primarily based on duration. For example, the German vowels [ɪ] and [i:] differ in terms of duration and their spectral properties. And Estonian distinguishes short, long, and very long vowels. And in English, tense vowels tend to have a longer steady-state portion than lax vowels. The steady-state portion is the stretch of the vowel during which the formant frequencies are relatively stable, usually toward the middle of the vowel, away from the influence from surrounding segments. In English, [æ] is an exception, in that it is a lax vowel (it does not occur in open syllables) but has a long duration. It has been observed

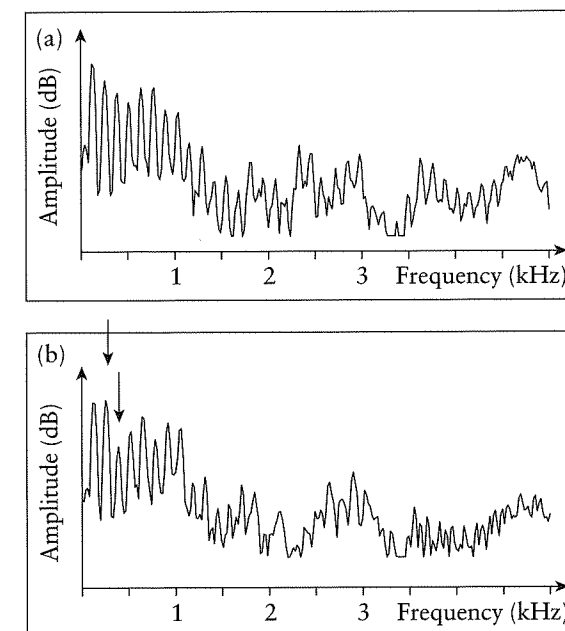


Figure 10.4 FFT spectra of (a) the oral vowel [a] and (b) the nasal vowel [ã] spoken by the same speaker as in Figure 10.3. The arrows in (b) indicate the increase in energy around 200 Hz and decrease in energy around 450 Hz.

that when tense vowels are flanked by consonants, the formant transitions into and out of tense vowels are relatively symmetrical while lax vowels have slightly longer transitions into, and much longer transitions out of, the vowel. In general, though, the primary difference between the English tense and lax vowels is one in terms of quality (formant pattern) rather than duration. A shortened [i] still sounds like an [i] just as a lengthened [ɪ] still sounds like an [ɪ], which is why, as mentioned in Section 3.2, the symbols [i] and [ɪ] are more appropriate to represent the tense and lax vowel in English, respectively, than the symbols [i:] and [i] or [ɪ:] and [ɪ].

Diphthongs (see Sections 2.4 and 3.2) are characterized by a change in formant pattern due to a change in articulatory shape during their production. This changing formant pattern primarily involves the first two formants and is very noticeable on spectrograms. Figure 10.5 shows spectrograms of the English diphthongs [aɪ], [ɔɪ], and [aʊ]. Acoustically, diphthongs are usually described by specifying their onset and offset frequencies. Notice that the formant frequency values at offset match those of the corresponding monophthongs ([ɪ, ʊ]) shown in Figure 10.2, as do the values at onset for [ɔ]. The onset of the diphthongs [aɪ] and [aʊ] is a vowel that is more central than the monophthong [a], with F2 onset values that are slightly higher.

10.2 Consonants

All consonants involve a modification of the airstream, ranging from mild in the case of approximants to severe in the case of plosives. Our overview of acoustic characteristics of consonants is organized in terms of manner of articulation, beginning with those of approximants, since they are most similar to those of vowels, and ending with those of plosives.

10.2.1 (Central) Approximants

In the production of approximants, two articulators approach each other without severely impeding the flow of air. The acoustic properties of approximants are therefore quite similar to those of vowels produced at a comparable location in the vocal tract. Their formant pattern is clear but somewhat weaker than for the vowels because of the approximants' slightly greater constriction, which results in a shorter steady-state portion and lower acoustic energy. For example, the articulation of the palatal approximant [j], in which the blade of the tongue is positioned relatively close to the palate, is quite similar to that of the vowel [i], the main difference being that the blade is closer to the palate for [j] than for [i]. The frequency spectrum of [j] therefore resembles that of [i], except that [j] is less

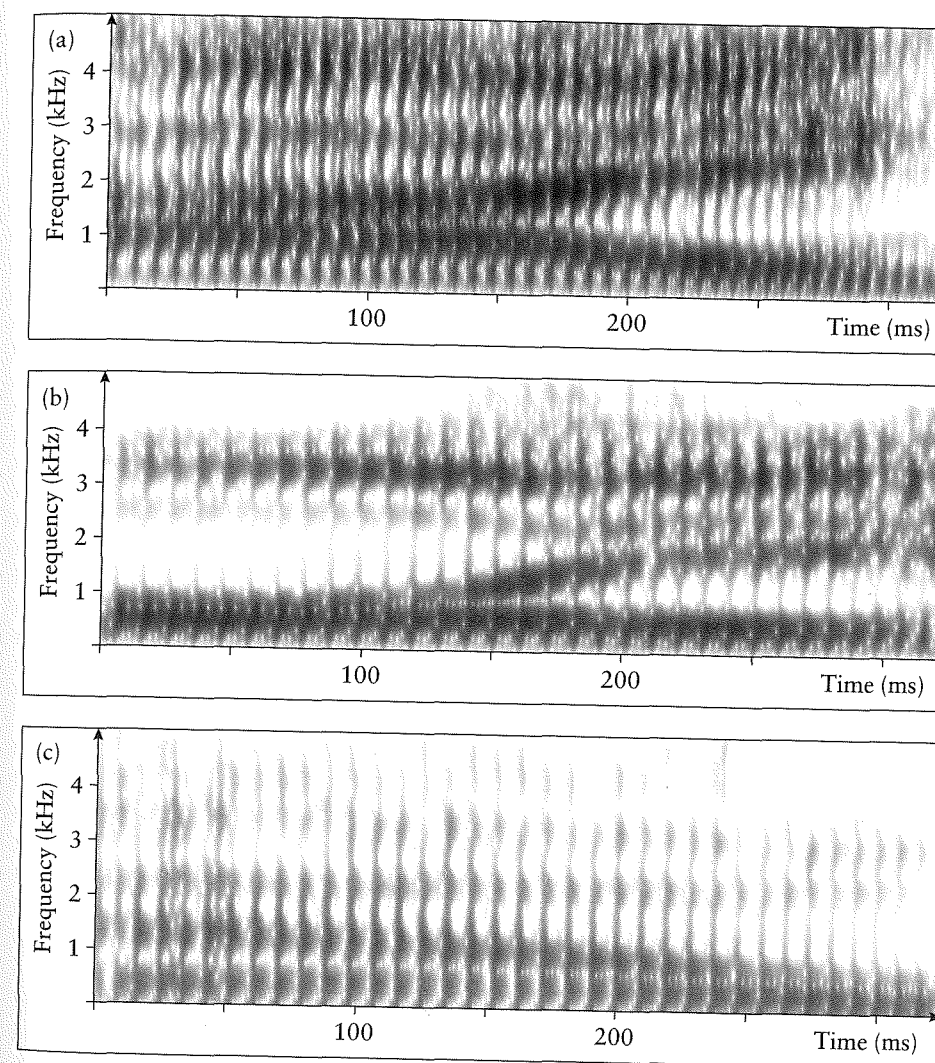


Figure 10.5 Spectrograms of the English diphthongs (a) [aɪ], (b) [ɔɪ], and (c) [aʊ] spoken by a male speaker.

intense since the airflow is impeded more than for [i]. Figure 10.6a shows a spectrogram for the utterance [qja]. The palatal approximant is characterized by a low F1, and high F2 and F3, similar to the vowel [i]. The transitions into and out of the approximant are quite pronounced, both in terms of frequency range and duration. Of course, the extent of the transition in the frequency domain depends on the vowel context. As illustrated in Figure 10.6b, when [j] is flanked by vowels with a similar formant pattern, as in [iji], the transitions are small.

During the production of the labial-velar approximant [w], the back of the tongue is brought closer to the velum and the lips are rounded. It is in that sense quite

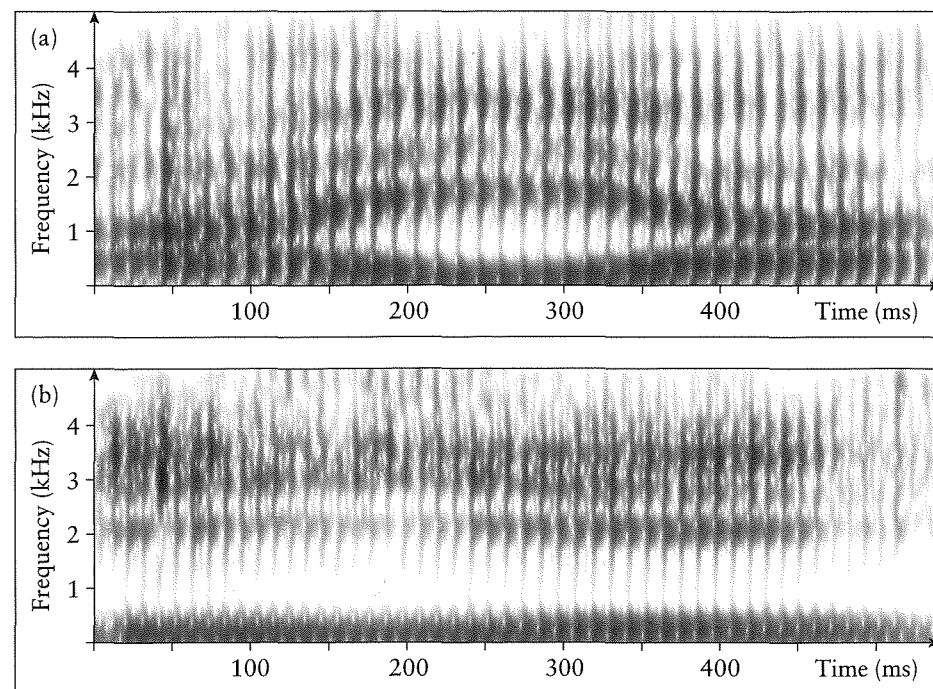


Figure 10.6 Spectrogram of the utterances (a) [aja] and (b) [iji] spoken by a male native speaker of English. Note that F3 is almost “on top of” F2 during the consonant.

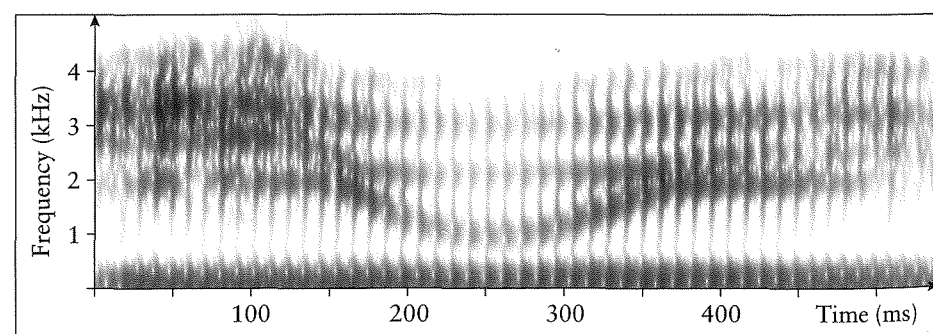


Figure 10.7 Spectrogram of the utterance [iwi] spoken by a male native speaker of English.

similar to the production of the vowel [u]. Figure 10.7 shows a spectrogram for the utterance [iwi]. During the labial-velar approximant, F1 and F2 are low and close together while F3 remains relatively steady at approximately 2,300 Hz, similar to the vowel [u]. Again, formant transitions depend on vowel context.

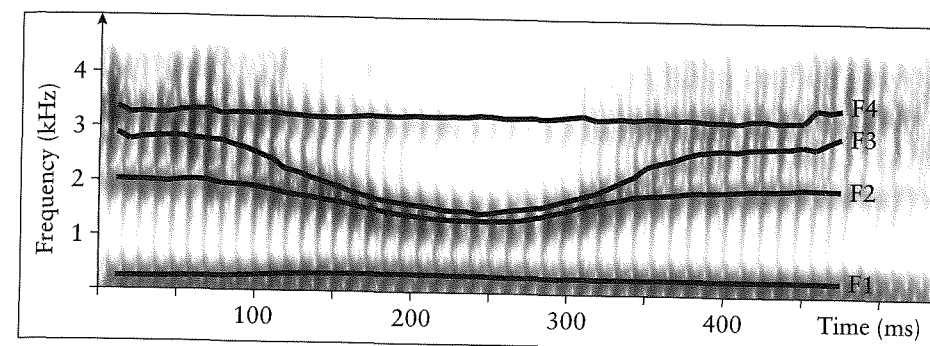


Figure 10.8 Spectrogram of the utterance [ɪɪ] spoken by a female native speaker of English. Smoothed lines have been superimposed to show the formant trajectories.

The retroflex approximant [ɭ] is typically produced with three simultaneous constrictions, all of which serve to lower F3: lip rounding, a narrowing near the alveolar ridge involving the tip of the tongue, and a narrowing in the pharynx because of retraction of the tongue root. As a result, [ɭ] is characterized by a very low third formant frequency, which comes close to F2. This is illustrated in Figure 10.8, where F3 during the approximant is around 1,800 Hz. In fact, [ɭ] has the lowest F3 of all English speech sounds, often dipping below 2,000 Hz even for female speakers.

In summary, approximants are distinct by being shorter and weaker than vowels and by having longer formant transition durations than vowels.

10.2.2 Fricatives

Fricatives are produced with a narrow constriction in the oral cavity. The turbulence resulting from air passing through this constriction (the position of which depends on the particular fricative) is the typical sound source for all fricatives (in addition to the voice source at the glottis for voiced fricatives). Fricatives can be characterized in terms of four attributes: spectral properties of the friction noise, amplitude of the noise, duration of the noise, and spectral properties of the transition into and out of the surrounding vowels.

The overall spectral shape of each fricative is mostly determined by the size and shape of the oral cavity in front of the constriction. The source-filter theory of speech production, introduced in Chapter 9, and illustrated by means of the acoustic characteristics of vowels, also explains the acoustic characteristics of consonants. Since obstruents are produced with a severe constriction, their resonant frequencies can be calculated for the front and back cavities separately, without having to consider complex interactions between the two. The simplest model for representing the articulatory configuration of these consonants is shown in



Figure 10.9 Vocal tract model representing the articulatory configuration for obstruents by means of three tubes corresponding, from left to right, to the back cavity, constriction, and front cavity, respectively

Figure 10.9. It requires three tubes, one for the front cavity, one for the constriction itself, and one for the back cavity.

The three-tube model consists of a combination of quarter-wavelength and half-wavelength resonators (see Section 9.1.1). The back cavity can be considered closed at both ends: one end is the glottis; the other end is the constriction in the oral cavity, which makes it a half-wavelength resonator. The short tube representing the constriction is also a half-wavelength resonator since it is open at both ends: both openings are large relative to the small area of the constriction. Finally, the front cavity is closed at one end at the constriction and open at the lips, making it a quarter-wavelength resonator.

We will use this model to calculate the formant frequencies. We will not consider the first formant here, which results from a **Helmholtz resonator**, a resonant system formed when a relatively wide back cavity is completely closed at one end (the glottis) and has a narrow constriction at the other end. The resonant frequency of this Helmholtz resonator is determined by the relative volumes of air in the back cavity and constriction, and is typically very low (below 1 kHz).

As an example, consider the alveolar fricative [s]. Based on X-ray and MRI studies of the vocal tract, we will assume a total vocal tract length of 16 cm, a back cavity length of 11 cm, a constriction length of 3 cm, and a front cavity length of 2 cm. Since the back cavity is a half-wavelength resonator, we will use the formula from Section 9.1.1:

$$f_k = k \times \frac{c}{2l}.$$

Using $l = 11 \text{ cm} = 0.11 \text{ m}$ and 340 m/s for the speed of the sound, this tube model predicts the following back cavity resonances:

$$f_k = k \times \frac{340 \text{ m/s}}{2 \times 0.11 \text{ m}} = k \times \frac{340}{0.22 \text{ s}} = k \times 1,545 \text{ Hz},$$

which yields 1,545 Hz, 3,090 Hz, 4,635 Hz, 6,180 Hz (for $k = 1, 2, 3, 4$), etc., as resonance frequencies.

For the front cavity, a quarter-wavelength resonator, we will use the formula from Section 9.1.1:

$$f_k = (2k - 1) \times \frac{c}{4l}.$$

Using $l = 2 \text{ cm} = 0.02 \text{ m}$, the following front cavity resonances are predicted:

$$f_k = (2k - 1) \times \frac{340 \text{ m/s}}{4 \times 0.02 \text{ m}} = (2k - 1) \times \frac{340}{0.08 \text{ s}} = (2k - 1) \times 4,250 \text{ Hz},$$

which yields 4,250 Hz, 12,750 Hz (for $k = 1, 2$), and so on, as resonance frequencies.

When assigning formant frequency numbers, the lowest resonance frequency that we calculated for fricatives is labeled F2 (F1 is from the Helmholtz resonator), the next one F3, etc. Thus, for [s], this model predicts F2 to be 1,545 Hz, F3: 3,090 Hz, F4: 4,250 Hz, F5: 4,635 Hz, etc. However, it is important to know whether a given formant frequency is associated with the back or the front cavity. This is because obstruent spectra are dominated by front cavity resonances. Friction noise or burst friction generated at the constriction primarily excites the cavity in front of the constriction. Since our calculations above showed that F2 and F3 are back cavity resonances, their influence is not as great as that of F4, which is a front cavity resonance. Thus, we expect the spectrum of [s] to show a concentration of energy at (and above) 4,250 Hz.

In general, it is the case that the longer the anterior cavity, the more distinct the resulting spectrum is since it operates as a resonating "chamber," which contributes more when it is larger. In addition, the spectral shape is more pronounced for the fricatives [s, z, ʃ, ʒ] because the airstream also hits the teeth. As a result, the alveolar and postalveolar fricatives are characterized by clear, distinct spectral shapes while labiodental and (inter)dental fricatives display a relatively flat spectrum. In particular, [ʃ, ʒ] typically exhibit a mid-frequency spectral peak around 2,500–3,500 Hz. Alveolar [s, z] are produced with a shorter anterior cavity than [ʃ, ʒ] and therefore display a primary spectral peak at higher frequencies, ranging from 4,000 to 7,000 Hz. In addition, since for these four fricatives the airstream hits the teeth, the high-frequency turbulence is very intense. In contrast, both [f, v] and [θ, ð] are characterized by a relatively flat spectrum, although recent research suggests that labiodentals may have a slightly higher spectral peak frequency (close to 8,000 Hz) than interdentals.

For more posterior constrictions (and longer front cavities), such as velar, uvular, and pharyngeal fricatives, the dominant energy is concentrated at lower frequencies, in the area of the first two formants. Velar fricatives have a concentration of energy in the region corresponding to F2 of the adjacent vowel and very little energy in the higher frequencies. The primary difference between

uvular and pharyngeal fricatives seems to be in terms of the first formant frequency. Consistent with their more posterior constriction, pharyngeal fricatives have a higher F1 than uvular fricatives.

Voiced fricatives have two energy sources: in addition to the turbulent noise generated by the constriction, the vibrating vocal folds provide low-frequency energy. Spectrograms of the voiced fricatives are similar to those of their voiceless counterparts, except that they contain additional low-frequency energy corresponding to vocal fold vibration and slightly less intensity in the higher frequencies because part of the energy of the airstream serves to make the vocal folds vibrate.

In terms of amplitude of the friction noise, [s, ʃ] have a substantially greater (10–15 dB) amplitude than [f, θ], which is why [s, ʃ] and their voiced counterparts [z, ʒ] are sometimes referred to as strong or sibilant fricatives in contrast to the weak or non-sibilant fricatives [f, θ, v, ð]. Within each of these two groups, measures of **relative amplitude** distinguish all four places of articulation. Relative amplitude is the difference between amplitude of the fricative and that of the following vowel measured in a specific frequency range (corresponding to F3 for alveolars and postalveolars and to F5 for labiodentals and interdental). This relative measure of amplitude indicates that postalveolars have a concentration of energy in the region corresponding to F3 of the following vowel while alveolars have their major energy in a much higher region. In addition, for the non-sibilants, fricative amplitude in the F5 region is greater for the labiodentals than the interdental.

Noise duration also serves to distinguish sibilant from non-sibilant fricatives, with [s, ʃ, ʒ] being longer than [f, θ]. Noise duration also distinguishes voiced and voiceless fricatives, with voiceless fricatives having longer noise durations than voiced fricatives.

Finally, it has been suggested that the formant transitions between fricative and vowel, especially the second formant transition, may cue place of articulation. Recent research indicates that F2 at the onset of the vowel tends to increase as place of articulation moves further back in the oral cavity.

In sum, fricatives have a relatively long noise portion with reasonably stable acoustic characteristics.

10.2.3 Plosives

During the production of plosives, the airstream is momentarily blocked by a complete constriction in the oral cavity. As discussed in Section 2.3.2, air pressure builds up behind the constriction. The release of the constriction and the subsequent movement of the articulators toward the next speech sound result in two primary cues to the place of articulation of the plosive, burst frequency and formant transitions. Figure 10.10 provides spectrograms of intervocalic plosives and illustrates a number of their distinctive acoustic characteristics. The closure interval between the initial vowel and the release burst of the plosive is a primary

characteristic of plosives and is of course due to the complete constriction in the vocal tract. This “gap” in the spectrogram corresponds to what is known as the closure portion of the consonant and contains no energy in the case of the voiceless plosives and only very low-frequency energy (called **voice bar**) in the case of fully voiced plosives.

Burst frequency – that is, the frequency spectrum of the release burst – varies as a function of the length of the vocal tract in front of the constriction, much as the dominant frequency of the friction noise for fricatives. Burst frequencies can be estimated with the same formulas used for the calculation of resonances for fricatives in the preceding sections, with a few modifications. Bilabials are formed at the lips without any appreciable portion of the vocal tract in front of the constriction. Since any constriction at the lips typically lowers the formant frequencies (see Section 9.1.2), bilabial bursts tend to have a concentration of energy in the lower frequencies (500–1,500 Hz). The short portion of the vocal tract in front of the alveolar constriction results in relatively higher burst frequencies for alveolars, ranging from about 2,500 to 4,000 Hz. Velars are characterized by a lower burst frequency than alveolars since the portion of the vocal tract in front of the velar constriction is substantially longer. The burst frequency for velars usually lies in between that of bilabials and alveolars, ranging from approximately 1,500 to 2,500 Hz. Release bursts are clearly visible in Figure 10.10 and their frequency can be seen to match our general description. Little is known about the acoustic differences between velar and uvular plosives. Consistent with their more posterior articulation, uvular plosives have a lower burst frequency.

Transitions of the second and third formants from the initial vowel into the closure gap and from the release burst into the following vowel can also provide cues to place of articulation. Formant transitions are the acoustic consequences of articulatory movement between adjacent vowels and plosives and are typically completed within 50 ms. All plosives share a low F1 because F1 during the constriction is at about 200 Hz. Since bilabials are characterized by low-frequency energy during the release, F2 and F3 transitions from vowel to plosive may descend into the closure gap and rise from the low-frequency burst toward the formant frequencies of the next vowel. Analogously, since alveolars typically have high-frequency energy during the release, F2 and F3 of the preceding vowel may rise into the closure gap and fall from the high-frequency burst into the following vowel. Finally, velar plosives with their mid-frequency bursts are often found to have an F2 and F3 that are close together in the mid-frequency range (2,000–2,500 Hz). That is, F2 rises and F3 falls into the closure portion; and F2 falls and F3 rises from the burst into the vowel. The proximity of F2 and F3 during the velar plosive is known as the **velar pinch**. While formant transitions provide cues to place of articulation, their extent is dependent on the quality of the adjacent vowel. For example, in the case of the sequence [ata], the second formant transition is much steeper (since [a] has a low F2) than in the sequence [iti] (since [i] has a high F2 that is closer to the high-frequency energy concentration of [t]).

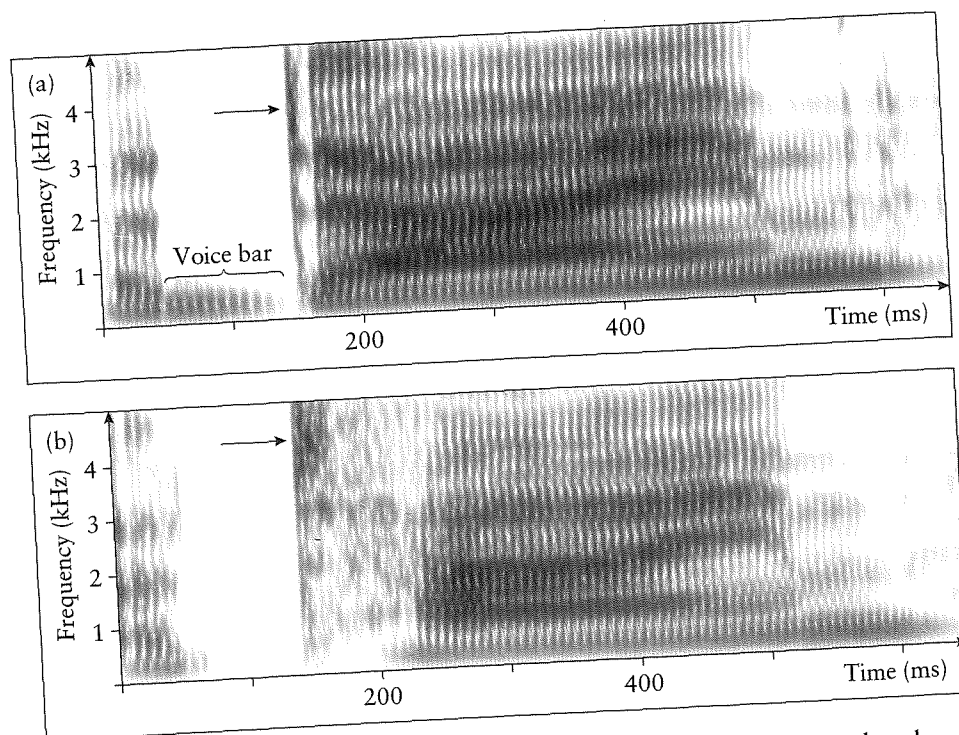


Figure 10.10 Spectrograms of the utterances (a) *a dime* with a voiced and (b) *a time* with a voiceless intervocalic plosive spoken by a female native speaker of American English. Arrows indicate the highest concentration of burst energy.

Finally, amplitude and duration provide information about consonant voicing but not place of articulation. A voiceless plosive in English typically has a longer closure duration, a stronger burst, and a longer (positive) VOT than a voiced plosive.

In sum, plosives are distinct by having a closure interval, a release burst, and rapidly changing formant transitions.

10.2.4 Nasals

For the production of nasal stop consonants the oral airflow is temporarily blocked by means of a complete constriction in the oral cavity. However, the velum is lowered such that air can escape only through the nasal cavity. The resulting sound is known as **nasal murmur**. Nasals require the use of two resonance cavities, oral and nasal, which are combined in a complex way. As a class, nasals are characterized by several properties. The first formant is very low because the total cavity length of all branches (pharyngeal, nasal, and oral) is longer than

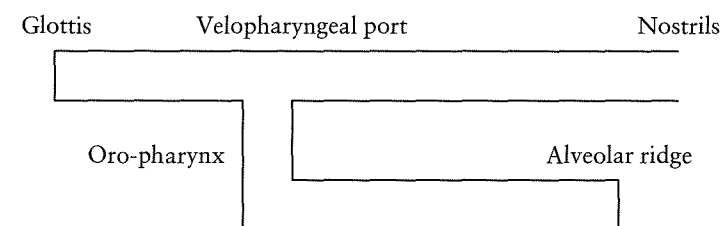


Figure 10.11 Schematic representation of the vocal tract configuration for an alveolar nasal stop consonant.

that for oral sounds. This low F1 is sometimes referred to as the **nasal formant**. The formants are weak (low in amplitude) because the airflow is impeded by a relatively narrow opening into the nasal cavity. The formants have an increased bandwidth, due to absorption of sound energy by the walls of both the oral and nasal cavities. And, finally, nasals are characterized by the presence of **anti-formants** due to the interaction between the oral and nasal cavities (see below).

For a given place of articulation, the oral cavity during the production of oral and nasal stops is very similar. For example, alveolar [d] and [n] both require a complete constriction at the alveolar ridge. As a result, the acoustic characteristics of [d] and [n] share certain properties. However, the only articulatory difference between the two sounds, the lowered velum for [n], turns out to have major acoustic consequences. Because of the open nasal cavity, the vocal tract needs to be modeled by means of a two-tube model (see Figure 10.11). One tube is closed-open and begins at the closed glottis and ends in the nasal cavity, which opens into the atmosphere. The other tube can be viewed as a side tube, open at one end – the uvula – and closed at the other – the point of constriction in the oral cavity.

Resonances of the oral cavity cannot be seen directly in a nasal's spectrum because the oral cavity has no opening to the atmosphere. Instead, the coupling or interaction between the oral and nasal cavities results in the presence of anti-formants. That is, those resonant frequencies in the nasal that are close to the resonances of the oral cavity are absorbed or canceled and "subtracted" from the overall spectrum because part of the acoustic energy of the pulsating airflow from the larynx causes resonance in the oral cavity. These anti-formants show up as conspicuous valleys in the spectrum, and their frequency varies as a function of place of articulation. The shorter the posterior part of the oral cavity – that is, the further back (more posterior) the place of articulation – the higher the resonances and, therefore, the higher the anti-formants in the output spectrum. In general, the anti-formant is located in the low-frequency range for bilabial [m] (750–1,250 Hz), in the mid-frequency range for alveolar [n] (1,450–2,200 Hz), and in the high-frequency range for velar [ŋ] (above 3,000 Hz). These characteristics are shown in Figure 10.12.

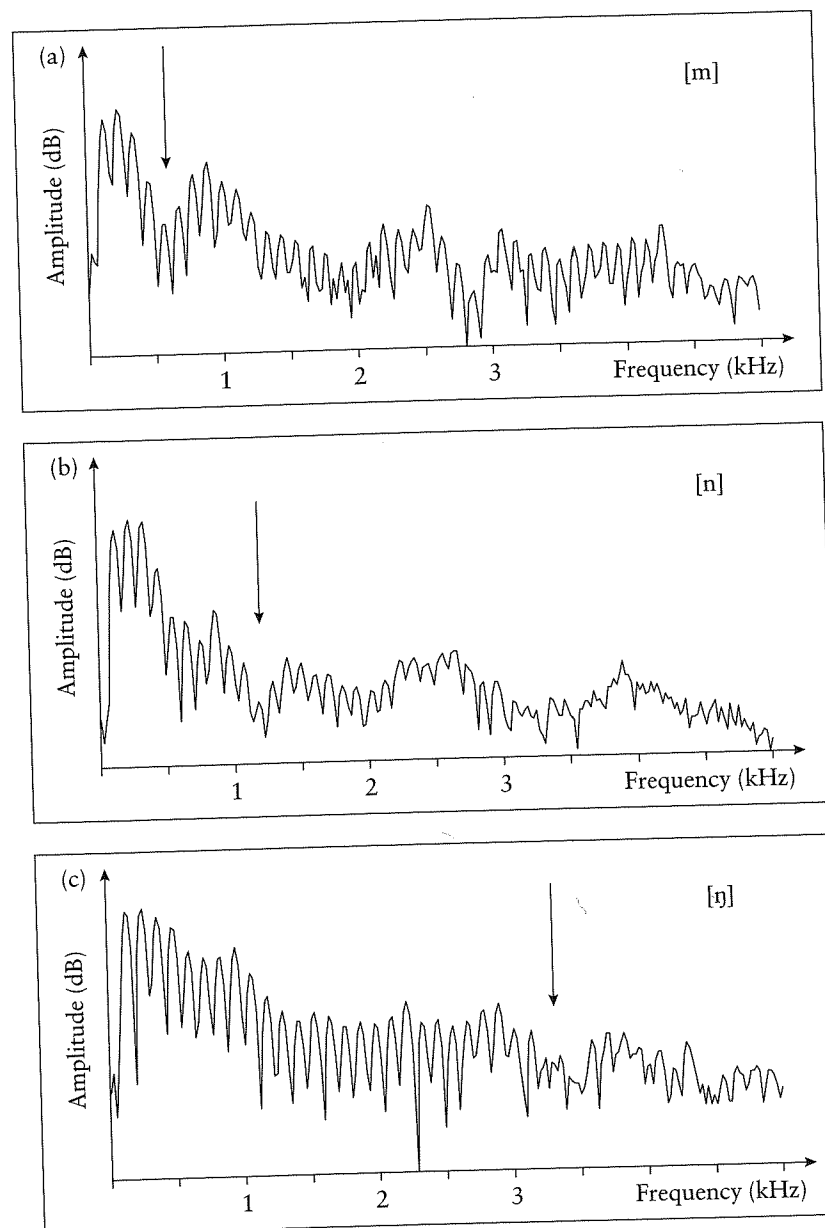


Figure 10.12 FFT spectra from the center of the nasal consonants in the utterances (a) [ama], (b) [ana], and (c) [aŋa]. The arrows indicate the locations of the anti-formants, which show up as dips in the spectra.

It would be unfair to conclude this section on cues to nasal place of articulation without mentioning that it is often very difficult to measure or even locate anti-formants. Moreover, individual variation in the size and shape of the nasal cavity increases the already substantial inter-speaker variability due to variation

in oral tract size. However, the formant transitions from preceding and into following segments may also contain information about place of articulation. The characteristics and interpretation of these transitions for nasal consonants are very similar to those for plosives produced at the same place of articulation (see Section 10.2.3).

10.2.5 Lateral approximants

Finally, the acoustic characteristics of the lateral alveolar approximant [l] are quite similar to those of nasals. In the production of [l], part of the tongue makes contact with the upper part of the mouth. Specifically, the tip of the tongue touches the alveolar ridge, while one or both sides of the tongue are lower and thus only approximate the roof of the mouth. [l] has a steady-state portion with a spectrum characterized by formants. Average adult male formant frequencies (F1 to F3) for [l] are approximately 340 Hz, 1,200 Hz, and 2,800 Hz, respectively. These characteristics are shown in Figure 10.13. For lateral alveolar approximants, a small pocket of air remains above the tongue body while air escapes on either side of the constriction. This situation is analogous to that for nasals in that the vocal tract can again be considered as consisting of a main tube and a side tube. The main tube extends from glottis to mouth opening while the pocket of air is modeled as a short side tube.

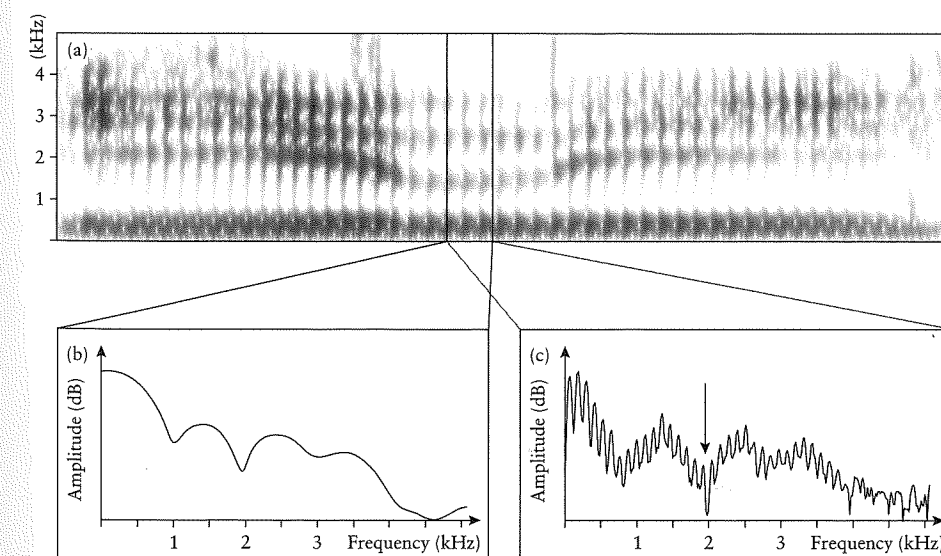


Figure 10.13 (a) Spectrogram of the utterance [ili] together with (b) LPC and (c) FFT spectra of the central part of [l], spoken by a male native speaker of English. The arrow in the FFT spectrum points to the anti-formant at 2,000 Hz.

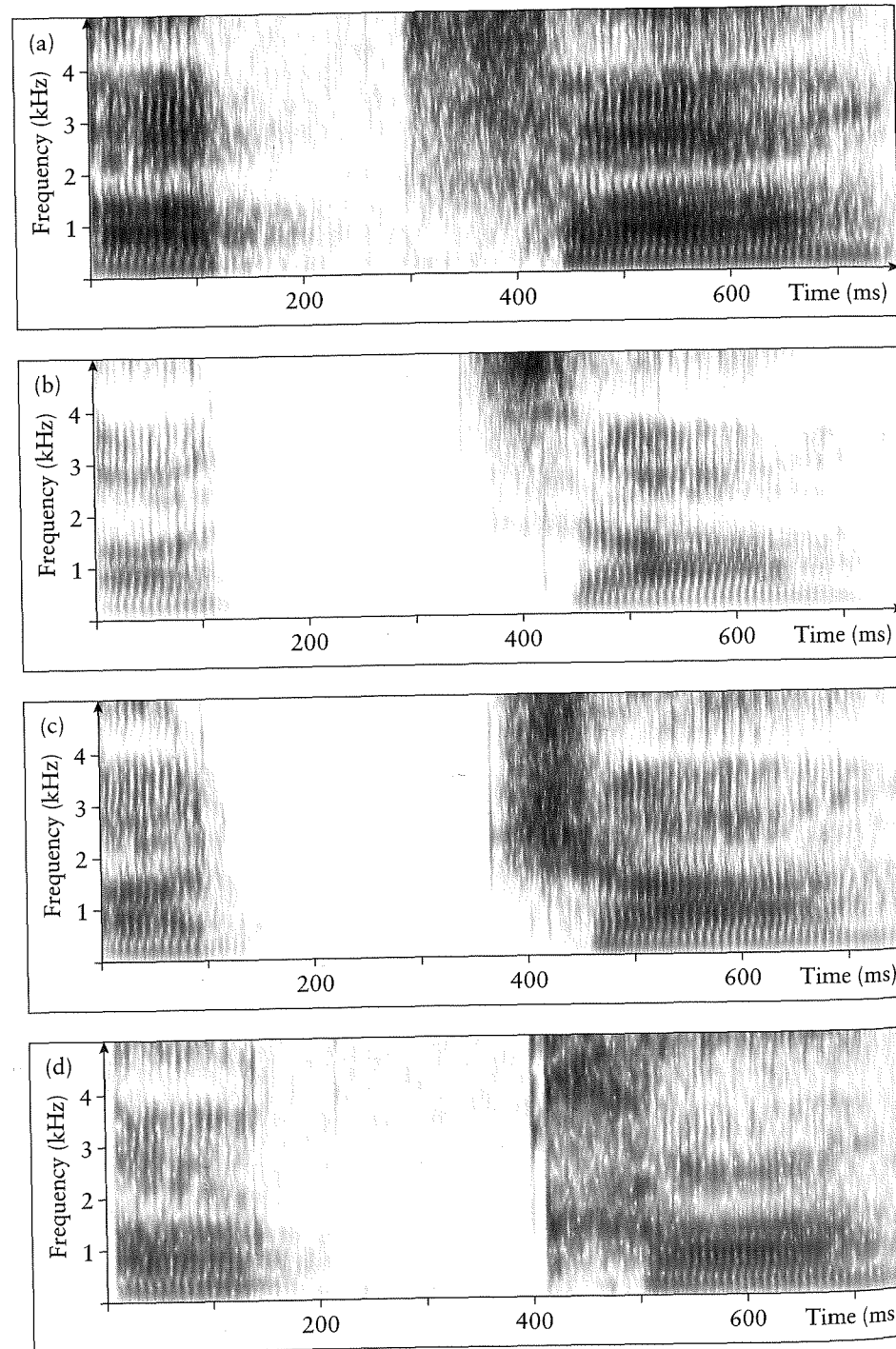


Figure 10.14 Spectrograms of (a) [apfa], (b) [at̪sa], (c) [at̪ja], and (d) [akxa] spoken by a Swiss German speaker.

The coupling between these two tubes again results in anti-formants. Formants from the side tube will cancel out formants from the main tube. Since the side tube is short, the anti-formant will be relatively high, at approximately 2,000–2,300 Hz for an adult male.

10.2.6 Affricates

Affricates combine aspects of plosive and fricative and their acoustic characteristics are similar to those of plosives and fricatives at the same place of articulation, except for the duration of the friction portion, which is shorter than for fricatives. Spectrograms of the utterances [apfa], [at̪sa], [at̪ja], and [akxa] are shown in Figure 10.14.

As a class, affricates can be distinguished from fricatives in terms of **rise time**, the temporal interval from the onset of the consonant until the maximum amplitude of the friction noise. Fricatives have a relatively long rise time since their amplitude builds up in a gradual manner. In contrast, affricates have a short rise time, reflecting the fact that following the plosive portion, the friction noise rapidly reaches maximum amplitude.

10.3 Summary

Our discussion of the acoustic characteristics of speech sounds suggests that it is easier to identify cues to manner of articulation and voicing than to place of articulation. In terms of manner of articulation, vowels and approximants stand out by the absence of any noise (burst or friction) portion. Vowels are distinct from approximants by having relatively intense and long steady-state portions. Plosives and affricates are both characterized by the presence of a closure gap without any (high-frequency) energy. Oral stops have a faster increase in energy following release than affricates, which in turn have a shorter rise time than fricatives. Finally, nasals are characterized by a very low first formant frequency and the presence of anti-formants.

Voiced speech sounds have low-frequency energy due to the vibrating vocal folds. For vowels, approximants, fricatives, and nasals, this voicing is usually present throughout the sound. For plosives, the voicing is most clearly observed during the closure portion.

For sounds with a substantial constriction, such as oral stops and fricatives, cues to place of articulation are largely determined by the length of the vocal tract in front of the constriction (i.e. the anterior part of the vocal tract). Therefore, oral stops and fricatives at the same place of articulation share certain acoustic properties. Bilabials are generally characterized by a predominance of energy in

the low-frequency range, alveolars by a predominance of high-frequency energy, while velars fall in between. Nasals are slightly different, in that the location of the characteristic anti-formant is determined by the length of the vocal tract between the uvula and oral constriction, which results in velars having the highest anti-formant frequency.

It should be kept in mind that while the acoustic properties discussed here are characteristic of the different manners of articulation, places of articulation, and voicing properties, they may not always be present for a given speech sound. The extent to which each or all of these properties occur depends on a variety of factors, including speaker, phonetic context, and speaking rate. This observation has led researchers to debate the extent to which the acoustic properties associated with a particular speech sound are invariant or context-dependent. The final section of this chapter is meant to provide a flavor of the kinds of data that researchers use to address the nature of the mapping between the physically continuous signal and discrete linguistic categories.

10.4 Variability and Invariance

As discussed earlier in this chapter, the most challenging aspect of consonants, particularly plosives, concerns cues to their place of articulation. While cues to plosive place include the frequency of the release burst and characteristics of the formant transitions, the role of each of these properties has been the topic of intense debate. Consider Figure 10.15.

This classic figure from Delattre et al. (1955) displays schematic spectrograms (of synthetic English speech) that are perceived by English listeners as [di] and [du], respectively. (Notice that these are simplified relative to natural speech tokens

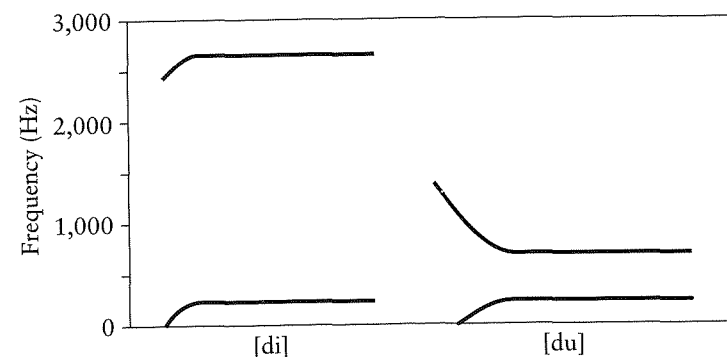


Figure 10.15 Stylized F1 and F2 pattern for [di] and [du] (after Delattre et al., 1955).

in that they do not have a release burst.) What is it in these two spectrograms that makes listeners perceive both syllables as starting with an alveolar plosive? Considering only [di] for the moment, the spectrogram can be decomposed into a number of components. The rising F1 transition indicates that it is a plosive. The location of the steady-state F1 and F2 indicates that the vowel is [i]. The only component left to indicate the place of articulation of the consonant is the rising F2 transition. However, consideration of [du] immediately suggests that a rising F2 transition is not likely to be a consistent cue to the alveolar place of articulation. While the rising F1 transition again indicates a plosive and the steady-state formants correspond to [u], the component that apparently indicates place of articulation is now a falling F2 transition in quite a different frequency range.

These kinds of observations led to the claim that most acoustic cues are highly *context-dependent*. According to this claim, there are no (or few) acoustic cues that are stable, consistent, or invariant. In this case, F2 transition as a cue to place of articulation varies drastically as a function of vowel context. That is, the same percept [d] is represented by two different acoustic cues, depending on the following vowel. These kinds of observations led Liberman and colleagues to claim that there is a “lack of invariance” in the speech signal, meaning that there is no simple one-to-one mapping between acoustic properties and phonetic percepts. Instead, different acoustic cues may yield the same phonetic percept (as in the [di–du] example).

It is important to note that this lack of invariance refers specifically to the acoustic domain. The issue is not whether or not there is invariance in the process of speech perception. Invariance clearly plays a role given that listeners map a variety of acoustic cues onto invariant phonological categories. Rather, the question is at which level this invariance resides. Demonstrations of the context-dependent nature of acoustic cues suggested that the invariance did not reside in the acoustic signal itself.

10.4.1 A Theory of acoustic invariance

In contrast to Liberman’s conclusion, other researchers (most notably, Stevens and Blumstein 1981) claim that invariant acoustic properties do indeed exist. One of the reasons why Blumstein and Stevens found invariant acoustic properties while Liberman and colleagues did not may have to do with how researchers analyzed the speech signal.

When Liberman started his quest for invariant acoustic properties, the best way to visualize and analyze the speech signal was the spectrogram. As discussed in Section 8.3.4, the spectrogram represents changes in frequency over time. When considering plosive-vowel syllables, three properties stand out on such spectrograms: release bursts, formant transitions, and formant steady-state portions. It is therefore tempting to treat each of these attributes as independent cues to consonant perception. Indeed, early studies in the 1950s did exactly that. For example,

the frequency location of the release burst was systematically manipulated to evaluate its role as a cue to place of articulation. The bursts were followed by combinations of steady-state two-formant patterns corresponding to different vowel qualities. Results failed to show a consistent mapping between burst frequency and perceived place of articulation. These kinds of findings reinforced the notion that the cues used in speech perception were highly context-dependent.

In their study of plosive-vowel syllables, Stevens and Blumstein used a spectral analysis technique that had not become generally available until the late 1960s, namely LPC analysis. As discussed in Section 8.3.5, in this analysis, spectral information is integrated over a time window. An LPC spectrum thus does not represent bursts, formant transitions, and steady states as separate events. Instead, the spectrum provides a single “snapshot” in which that information is integrated over time. The typical analysis window used in the study of plosive spectra is on the order of 10–25 ms. It is believed that this kind of representation is more like that derived by the auditory system.

Analysis of LPC spectra computed over only the first 25 ms of CV sequences suggested that the global shape of the spectra contained cues to place of articulation that were invariant across such sources of variability as phonetic context and vocal tract size. Templates representing the global shape associated with each place of articulation are shown in Figure 10.16. Bilabial consonants were found to be consistently associated with either a falling spectrum, indicating a predominance of energy in the lower frequencies, or a relatively flat spectrum without any clear concentration of energy. The rising spectrum for the alveolars reflected the predominance of energy in the higher frequencies, while the velar spectrum indicated that velars typically have a concentration of energy in the mid-frequency region, a spectral shape labeled “compact.”

LPC spectra were then computed for CV and VC syllables produced by a new set of speakers and the templates were used to classify the consonants in terms of place of articulation. Results showed that place of articulation was classified with 85 percent accuracy for 1,800 tokens produced by four male and two female speakers (Blumstein and Stevens 1979).

Subsequent research on invariance focused on a more *dynamic* cue, namely the change in distribution of high-frequency energy as compared to low-frequency energy between consonant onset and the onset of the following vowel. This approach was a refinement of the earlier research in that it still captured the basic notion that bilabials are characterized by a relative predominance of energy in the lower frequencies while alveolars showed a predominance of energy in the higher frequencies. Using this dynamic criterion, 91 percent of the labial, dental, and alveolar plosives in English, French, and Malayalam were correctly classified (Lahiri et al. 1984; see also Kewley-Port 1983 for a dynamic approach).

Integrated dynamic invariant acoustic cues have also been documented for place of articulation in nasals. This research focused on the spectral change surrounding the nasal-vowel boundary in nasal-vowel syllables. Spectra were computed for the last two glottal pulses of the nasal murmur and the first two of the release into the vowel. A metric was established that captured the relative energy changes

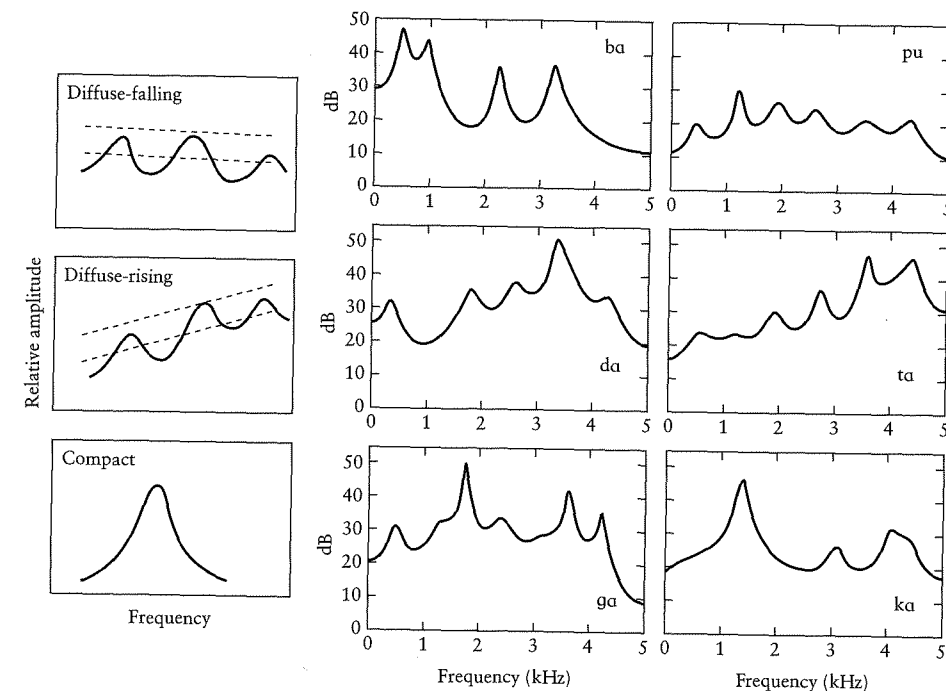


Figure 10.16 The middle and right columns show LPC spectra for labial, alveolar, and velar plosives, computed over the first 25 ms of CV syllables. The left column shows the diffuse-falling, diffuse-rising, and compact templates that capture the spectral shape of labial, alveolar, and velar plosives.

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in a low and high frequency range, corresponding to those reported as the location of the anti-formant for [m] and [n], respectively. This metric revealed that [m] was characterized by a rapid increase in energy in the lower frequencies while [n] showed a rapid increase in the higher frequencies. Classification of nasal place of articulation reached 89 percent accuracy across three speakers and five vowel contexts (Kurowski and Blumstein 1987).

Finally, invariant acoustic cues have also been reported for the manner distinction between plosives and approximants, specifically the glides [w] and [j]. In addition to differences in formant frequency onset values and formant transition duration, plosive-vowel and glide-vowel syllables such as [ba] and [wa] also differ in terms of the relative change in energy between the release of the consonant and the onset of voicing in the following vowel. Since plosives are produced with a complete constriction followed by a release, the change in energy from plosive to vowel is relatively large, certainly larger than that for approximants, which have only a moderate constriction. Taking the ratio of the energy in the first pitch period after the release and the energy in the period containing the release, Mack and

Blumstein (1983) found that plosives had consistently larger ratios than glides. Specifically, the adoption of a cut-off ratio of 1.37 allowed classification of consonants as either plosive or glide with 92 percent accuracy. In addition, the same metric distinguished [d, g] from [j] with 90 percent accuracy.

In sum, the studies summarized in this section suggest that invariant patterns corresponding to place and manner of articulation may be found in the acoustic signal. Crucially, these patterns do not focus on a single acoustic property, because, as Liberman and colleagues clearly showed, individual properties are usually highly context-dependent. Instead, dynamic cues that integrate several properties over a brief period seem to be much less affected by variations in speaker and phonetic context.

More recently, several statistical approaches to the classification of consonants in terms of place of articulation have been proposed. In an attempt to accommodate context effects, Forrest et al. (1988) used **spectral moments** to classify word-initial voiceless obstruents. To do this, a series of fast Fourier transforms (FFTs) was calculated every 10 ms from the onset of the obstruent. Each FFT was treated as a random probability distribution (i.e. as a set of values that can be statistically analyzed) from which the first four moments (mean, variance, skewness, and kurtosis) were computed. The mean of the frequency values weighted by their amplitude reflects *where* energy is concentrated on the frequency scale. The variance gives a measure for the range over which this energy is spread. The skewness refers to spectral tilt (the overall slant of the energy distribution), and the kurtosis is an indicator of the "peakedness" of the distribution. Using a spectral moments metric thus incorporates both local (spectral peak) and more global (spectral shape) information. The spectral moments were entered into a statistical discriminant analysis for classification in terms of place of articulation.¹ Correct classification of place based on the first 20 ms of the plosives (produced by five females and five males) averaged 89 percent.

A different statistical approach is exemplified by the use of **locus equations**. Locus equations provide a way of quantifying the fact first observed by Lindblom (1963) for voiced plosives that F2 frequency at the onset of the vowel following a voiced plosive is correlated with its frequency in the vowel midpoint of a plosive-vowel syllable. A locus equation can be defined as "a straight-line regression fit to coordinates formed by plotting onsets of F2 transitions in relation to their coarticulated F2 midvowel 'target frequencies'" (Sussman et al. 1993, p. 1256). In other words, each production of a CV syllable yields one datapoint consisting of an F2 value measured at vowel onset and an F2 value measured in the middle of the vowel. When all these points are plotted in an F2(onset) by F2(middle) space, a best-fitting line is drawn through this cloud of points such that each point is as close to this line as possible. Locus equations can be considered as indicators of the degree of coarticulation between a plosive and following consonant. If there were maximal coarticulation, the target F2 of the vowel would already be reached during the preceding plosive. The F2 at the middle of the vowel would therefore determine the F2 at the onset of the vowel and the locus equation would have a slope of 1, as illustrated in Figure 10.17. If, however, there were

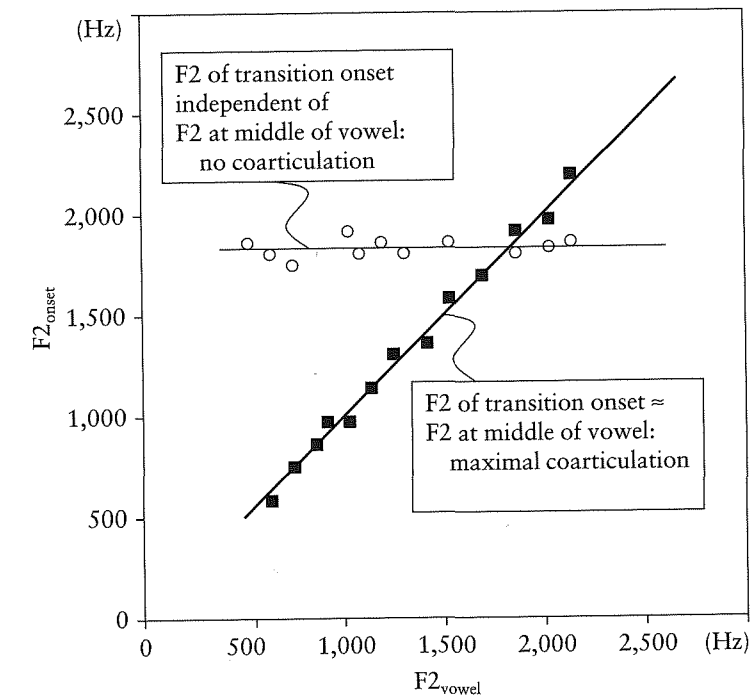


Figure 10.17 Schematic representation of locus equations for maximal and minimal consonant-to-vowel coarticulation.

no coarticulation, the F2 of the consonant would remain the same, regardless of the quality of the following vowel, and the locus equation would have a slope of 0, as shown in Figure 10.17.

Locus equations indicate that bilabials show the greatest degree of coarticulation, with steep slopes, velars an intermediate degree, and alveolars exhibit the smallest degree of coarticulation with relatively shallow slopes, as shown in Figure 10.18.

Locus equations constitute a dynamic representation of speech sounds since they express a relation between F2 at different points (vowel onset and midpoint) in the speech signal. In their study of English voiced plosive consonants, Sussman et al. (1991) found that discriminant analysis (based on the slope and y-intercept values) of each speaker's locus equations for each of the three places of articulation yielded a correct classification rate of 100 percent.

In conclusion, this chapter discussed the broad acoustic correlates of a number of sound classes. As shown on the spectrograms (e.g. Figure 10.6), some of these correlates vary depending on the context in which they occur. The debate about the existence and nature of invariant cues will likely continue for some time. While the classification rates of recent statistical metrics are very impressive, it remains to be determined whether human listeners base their identification of place

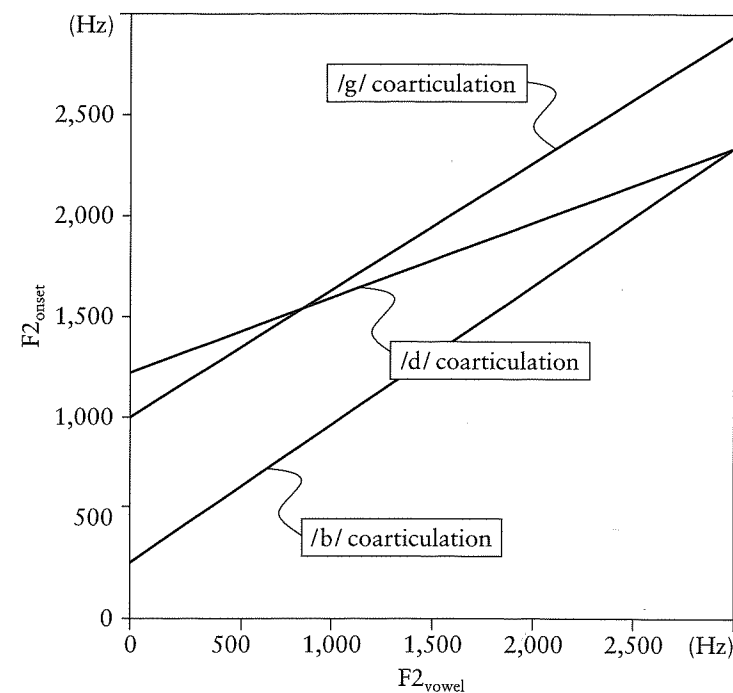


Figure 10.18 Regression lines of F2 transitions in relation to their F2 midvowel target frequencies for [p, t, k] and the three vowels [i, a, u] in German based on 10 speakers.

of articulation on similar sources of information. We will return to these issues in Chapter 13.

Exercises

- 1 Which formants are important for distinguishing the vowels in Dutch and German? What information does each formant provide?
- 2 How do oral and nasal vowels differ in spectrograms? What causes this difference?
- 3 How can monophthongs and diphthongs be differentiated in spectrograms?
- 4 What are four acoustic characteristics used to distinguish fricatives? In terms of these characteristics, how do [s, z, ʃ, ʒ] differ from [f, v, θ, ð]?
- 5 The acoustics of sounds with a substantial constriction can be modeled by means of a three-tube model consisting of a back cavity, a constriction, and a front cavity. Assume a model with a total tube length of 16 cm (13 cm for

the tubes, 3 cm for the constriction) and calculate F2, F3, and F4 and their cavity affiliations for the following consonants:

/s/ as in 'see' (length of the back cavity = 11 cm)

/ʃ/ as in 'she' (length of the back cavity = 10 cm)

In which frequency range do you expect the energy to be concentrated for each of these two consonants?

- 6 What acoustic characteristics are used to discern the place of articulation of stops? What indicates whether a stop is voiced or voiceless?
- 7 What is meant by the claim that the speech signal is characterized by a "lack of invariance"?

Note

- 1 A statistical discriminant analysis assigns cases to groups based on a number of (acoustic) parameters.